

ADVANCED COMPUTER NETWORKS

Complete Study Notes

Covers Full Syllabus • Key Points • Tables • Working Principles • Advantages & Limitations

MODULE 1: Basic Networking Concepts & Layering

► 1.1 What is a Network?

A computer network **connects multiple devices (nodes)** so they can share data and resources. Think of it as a postal system — just for digital data!

🔑 **KEY:** A network = Nodes (devices) + Links (cables/wireless) + Protocols (rules)

◆ Key Networking Terms

Term	Meaning	Example
Protocol	Rules for communication	TCP, HTTP, FTP
Bandwidth	Max data transfer rate	100 Mbps
Latency	Delay in data transmission	20 ms ping
Packet	Unit of data sent over network	1500 bytes Ethernet frame
Topology	Physical/logical layout	Star, Bus, Ring, Mesh
Node	Any device on network	PC, Router, Switch

► 1.2 The OSI Model (7 Layers)

The OSI model breaks networking into 7 layers — each layer has a specific job and talks to layers above/below it.

🔑 **KEY:** Mnemonic: 'Please Do Not Throw Sausage Pizza Away' (Physical → Application)

Layer #	Layer Name	Function	Protocol Examples
7	Application	User-facing services (email, web)	HTTP, FTP, SMTP, DNS
6	Presentation	Data format, encryption, compression	SSL/TLS, JPEG, ASCII
5	Session	Manages sessions/connections	NetBIOS, RPC

Layer #	Layer Name	Function	Protocol Examples
4	Transport	End-to-end delivery, error checking	TCP, UDP
3	Network	Routing packets across networks	IP, ICMP, OSPF
2	Data Link	Node-to-node transfer, MAC addressing	Ethernet, Wi-Fi (802.11)
1	Physical	Actual bits on wire/air	Cables, Radio waves, Fiber

► 1.3 TCP/IP Model (4 Layers) Transmission Control Protocol/Internet Protocol

The practical model used on the internet — simpler than OSI.

TCP/IP Layer	Maps to OSI Layers	Key Protocols
Application	5, 6, 7	HTTP, DNS, FTP, SMTP, SSH
Transport	4	TCP, UDP
Internet	3	IP, ICMP, ARP
Network Access	1, 2	Ethernet, Wi-Fi, PPP

MODULE 2: Link Layer

► 2.1 What is the Link Layer?

The Link Layer (Layer 2) handles data transfer between directly connected nodes. It deals with framing, MAC addresses, and error detection.

 **KEY:** Link Layer = Data Link Layer in OSI. Responsible for hop-to-hop delivery (one link at a time).

◆ Key Functions

- Framing – wraps data into frames
- MAC Addressing – unique hardware address for each device
- Error Detection – CRC (Cyclic Redundancy Check)
- Flow Control – prevents sender from overwhelming receiver
- Medium Access Control (MAC) – decides who talks when on shared medium

◆ MAC Protocols — How Devices Share the Channel

Protocol Type	How it Works	Example
TDMA	Time Division: each device gets fixed time slot	Cellular 2G

Protocol Type	How it Works	Example
FDMA	Frequency Division: each device gets frequency band	FM Radio
CSMA/CD	Listen before talk; detect collisions (wired)	Ethernet
CSMA/CA	Listen before talk; avoid collisions (wireless)	Wi-Fi 802.11
Token Ring	Pass token to control access	Old LANs

✓ Advantages	✗ Limitations
CSMA/CD works well on wired networks	CSMA/CA has overhead due to RTS/CTS
Low overhead with TDMA in cellular	TDMA wastes slots if device is idle
MAC protocols prevent data collisions	Hidden terminal problem in wireless

MODULE 3: Network Layer

► 3.1 Network Layer Overview

The Network Layer (Layer 3) is responsible for routing packets from source to destination across multiple networks using IP addresses.

🔑 **KEY:** Main job: Logical addressing (IP) + Routing packets across networks

◆ IP Addressing

Version	Address Size	Example	Addresses Available
IPv4	32-bit	192.168.1.1	~4.3 Billion
IPv6	128-bit	2001:0db8::1	340 Undecillion

► 3.2 Routing

Routing is the process of finding the best path for data packets to travel from source to destination.

◆ Types of Routing

Type	How it Works	Advantage	Limitation
Static	Admin manually sets routes	Simple, predictable	Doesn't adapt to changes
Dynamic	Routers exchange info and auto-update	Adapts to failures	More complex, overhead

Type	How it Works	Advantage	Limitation
Distance Vector	Each router knows distance to neighbors (RIP)	Easy to implement	Slow convergence
Link State	Each router knows entire topology (OSPF)	Fast convergence	More memory needed
Path Vector	BGP – used between ISPs	Policy-based routing	Complex configuration

◆ OSPF vs RIP vs BGP

Protocol	Full Name	Type	Used For	Metric
RIP	Routing Info Protocol	Distance Vector	Small networks	Hop count (max 15)
OSPF	Open Shortest Path First	Link State	Large enterprise	Cost (bandwidth)
BGP	Border Gateway Protocol	Path Vector	Internet/ISPs	Policy-based

🔑 **KEY:** Dijkstra's Algorithm is used in OSPF to find shortest paths. Bellman-Ford is used in RIP.

MODULE 4: End-to-End Layer (Transport Layer)

► 4.1 Transport Layer Functions

The Transport Layer provides communication between applications running on different hosts.

🔑 **KEY:** Two main protocols: TCP (reliable) and UDP (fast, unreliable)

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery	No guarantee
Order	In-order delivery	No ordering
Speed	Slower (overhead)	Faster (low overhead)
Use Case	Web, Email, File Transfer	Video streaming, DNS, Gaming
Header Size	20 bytes minimum	8 bytes

◆ TCP 3-Way Handshake

Step 1: Client sends SYN (synchronize) to server

Step 2: Server replies with SYN-ACK (synchronize + acknowledge)

Step 3: Client sends ACK (acknowledge) — connection established!

KEY: After handshake, data flows **bidirectionally**. Connection ends with FIN/ACK 4-way termination.

MODULE 5: Congestion Control

► 5.1 What is Congestion?

Congestion happens when too many packets overwhelm the network, **causing delays, packet loss, and reduced throughput**. Think of a traffic jam on a highway!

KEY: Congestion $\neq$ Congestion Control. Congestion is the problem; Congestion Control is the solution.

► 5.2 TCP Congestion Control Mechanisms

Mechanism	How it Works	When Used
Slow Start	Begin with small window; double it each RTT	At connection start or after timeout
Congestion Avoidance	Increase window linearly (additive increase)	When $cwnd \geq ssthresh$
Fast Retransmit	Retransmit after 3 duplicate ACKs	When packet loss detected quickly
Fast Recovery	Reduce window by half, don't go to slow start	After fast retransmit
AIMD	Additive Increase, Multiplicative Decrease	Standard TCP behavior

► 5.3 TCP Variants

Variant	Key Feature	Best For
TCP Tahoe	Slow start + fast retransmit	Basic reliable transport
TCP Reno	+ Fast Recovery	Most common variant
TCP NewReno	Improved fast recovery for multiple losses	Multiple packet loss scenarios
TCP Vegas	RTT-based congestion detection (proactive)	Low-latency links
TCP CUBIC	Cubic growth function for high-bandwidth	Modern Linux default

Variant	Key Feature	Best For
TCP BBR	Model-based; maximizes bandwidth	Google's modern networks
TCP SACK	Selective ACK — only re-send lost packets	High-loss networks

► 5.4 Active Queue Management (AQM)

AQM proactively drops/marks packets before queues are full, giving early congestion signals.

AQM Scheme	Full Name	Key Idea
RED	Random Early Detection	Randomly drop packets when avg queue grows
WRED	Weighted RED	Different thresholds per traffic class
CoDel	Controlled Delay	Control sojourn time, not queue length
FQ-CoDel	Fair Queue CoDel	Per-flow queuing + CoDel — used in Linux

Sojourn time is the total duration a unit spends within a specific system spanning from entry to departure

MODULE 6: Network Traffic Modeling & Measurement

► 6.1 Traffic Models

Traffic models help predict and analyze network behavior for planning and design.

Model	Description	Used For
Poisson	Random arrivals at constant rate	Traditional telephony
Self-Similar / Fractal	Bursty — same pattern at all scales (Hurst parameter)	Internet traffic, LAN
Markov Models	State-based probabilistic model	Packet loss modeling
MMPP	Markov Modulated Poisson Process	Variable rate traffic bursts

🔑 **KEY:** Internet traffic is NOT Poisson — it's self-similar and bursty (proven by Willinger et al., 1995)

► 6.2 Network Measurement Techniques

- Active Measurement – send probe packets and measure (e.g., ping, traceroute) - ICMP
- Passive Measurement – monitor traffic without injecting packets (e.g., packet sniffing)
- SNMP – Simple Network Management Protocol for device monitoring
- NetFlow/IPFIX – flow-level traffic data collection

► 6.3 Simulation Issues

Traffic models in computer networks describe how data (packets) is generated and flows over time. These are important for performance analysis, congestion control, and network design—and GATE often asks conceptual + numerical questions from here.

1. Poisson Traffic Model (Most Important)

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Idea

- Packets arrive randomly but with a constant average rate (λ)
- Independent arrivals

Key Points

- Inter-arrival time $\rightarrow$ **Exponential distribution**
- Used in:
 - Telephone networks
 - Packet-switched networks

Example

- If $\lambda = 5$ packets/sec $\rightarrow$ on average 5 packets arrive every second, but exact arrival is random

2. Constant Bit Rate (CBR)

Idea

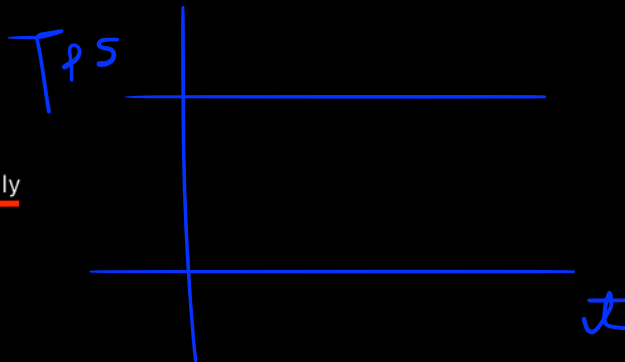
- Traffic is generated at a **fixed rate**

Example

- Sending 64 kbps voice stream continuously

Features

- ✓ Predictable
- ✓ No burst
- ✗ Not realistic for most data traffic



3. Bursty Traffic Model

Idea

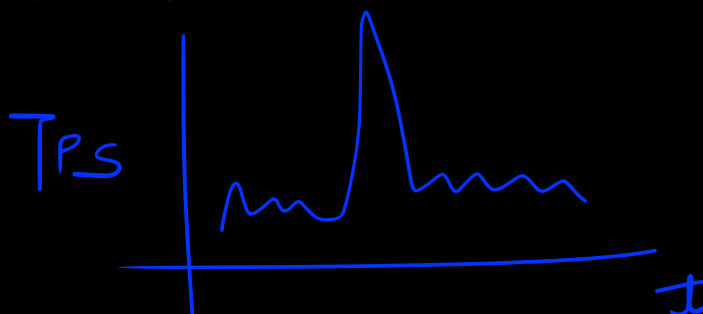
- Traffic comes in **bursts (sudden spikes)** followed by idle time

Example

- Web browsing:
 - Load page $\rightarrow$ burst of packets
 - Reading $\rightarrow$ no traffic

Features

- ✓ Realistic model
- ✗ Hard to predict



4. Self-Similar Traffic (Fractal Traffic)

◆ Idea

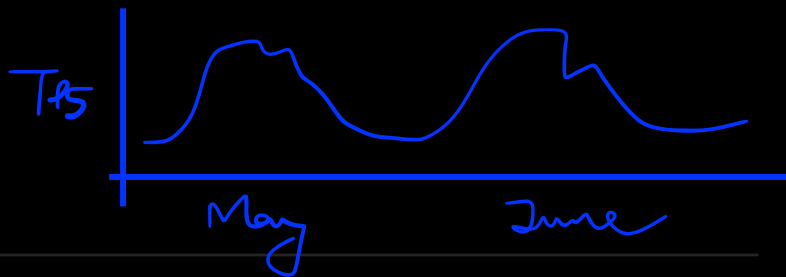
- Traffic pattern looks similar at different time scales

◆ Example

- Internet traffic (LAN, WAN)

◆ Features

- ✓ Captures real-world behavior
- ✓ Long-range dependence



5. ON-OFF Model

◆ Idea

- Traffic alternates between:
 - **ON state** → packets sent
 - **OFF state** → no packets

◆ Example

- Video streaming / VoIP

6. Fluid Flow Model

◆ Idea

- Traffic treated as a continuous flow (like water) instead of discrete packets

◆ Use

- Used in mathematical analysis of congestion

7. Markov Modulated Traffic (Advanced)

◆ Idea

- Traffic behavior changes based on states (Markov process)

◆ Example

- Network switches between low load and high load states



Quick Comparison (GATE Revision)

Model	Nature	Realism	Key Use
Poisson	Random	Medium	Analysis
CBR	Constant	Low	Voice
Bursty	Irregular	High	Web traffic
Self-similar	Fractal	Very High	Internet
ON-OFF	Alternating	High	Streaming
Fluid	Continuous	Medium	Theory

Network simulation is used to test designs before deployment. Key simulators: NS-2, NS-3, OPNET, GNS3.

Issue	Description	Solution
Scalability	Simulating millions of nodes is too slow	Use emulation or hybrid models
Accuracy	Models may not match real traffic	Use real traffic traces
Reproducibility	Random events make results vary	Fix random seeds
Validation	Sim results $\neq$ real results	Always validate with testbeds

MODULE 7: Routing & Router Design, QoS

► 7.1 Router Architecture

A modern router has 4 main components:

Component	Function
Input Ports	Receive packets, lookup forwarding table, classify
Switching Fabric	Move packets from input to output ports (crossbar/bus/shared memory)
Output Ports	Buffer and schedule packets for transmission
Routing Processor	Run routing protocols, update forwarding table

► 7.2 Scheduling & QoS

Quality of Service (QoS) ensures important traffic (voice, video) gets priority over bulk data.

🔑 **KEY: QoS Guarantees:** Bandwidth, Delay, Jitter (delay variation), Packet Loss Rate

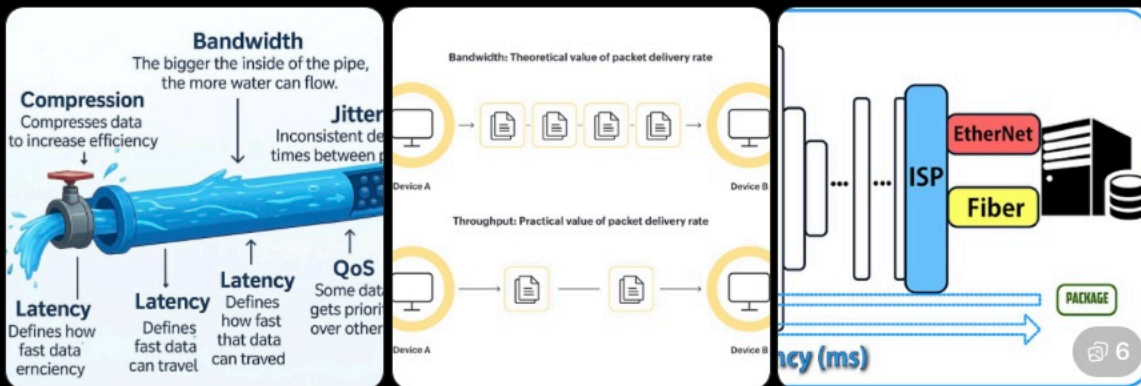
◆ Scheduling Algorithms

Algorithm	How It Works	Best For
FIFO/FCFS	First in, first out — no priority	Simple best-effort
Priority Queuing	High priority packets go first always	Time-sensitive traffic
Round Robin (RR)	Serve each queue in turn equally	Fair sharing
WFQ	Weighted Fair Queuing — give proportional shares	Multi-class traffic
WRR	Weighted Round Robin — simpler than WFQ	ISP traffic engineering



Quality of Service (QoS) defines how well a network delivers traffic, especially for real-time apps like video calls, gaming, VoIP, etc. The four core guarantees you mentioned are the backbone of QoS questions in GATE.

🎯 1. Bandwidth (Throughput Guarantee)



💡 Idea

- Maximum data rate a network can provide (bits/sec)

💡 Example

- 10 Mbps link → can send 10 million bits per second

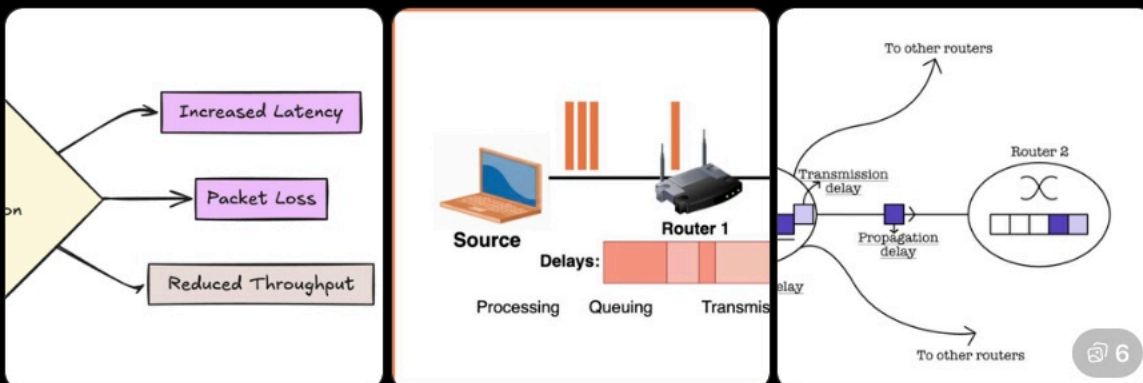
💡 Why important?

- Video streaming needs **high bandwidth**
- Low bandwidth → buffering

💡 GATE Insight

- Bandwidth $\neq$ actual throughput (due to delay, loss, congestion)

🕒 2. Delay (Latency Guarantee)



◆ Idea

- Time taken for a packet to go from source → destination

◆ Total Delay Components

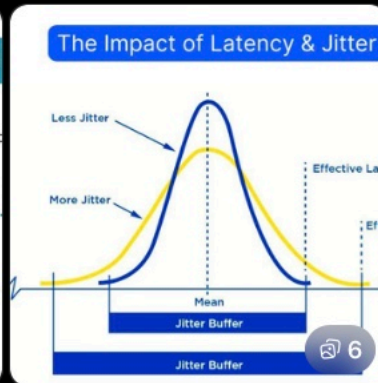
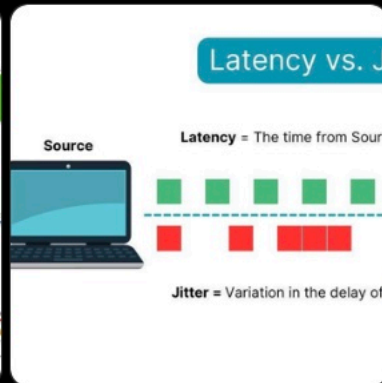
$$\text{Total Delay} = \text{Transmission} + \text{Propagation} + \text{Queuing} + \text{Processing}$$

- **Transmission delay** → packet size / bandwidth
- **Propagation delay** → distance / speed of signal
- **Queuing delay** → waiting in buffer
- **Processing delay** → router processing

◆ Example

- Online gaming → needs **very low delay**

3. Jitter (Delay Variation)



Idea

- Variation in packet delay over time

If packets arrive like:

- 10 ms, 12 ms, 11 ms → low jitter ✓
- 10 ms, 50 ms, 5 ms → high jitter ✗

Why important?

- Critical for:
 - VoIP
 - Video calls

Fix

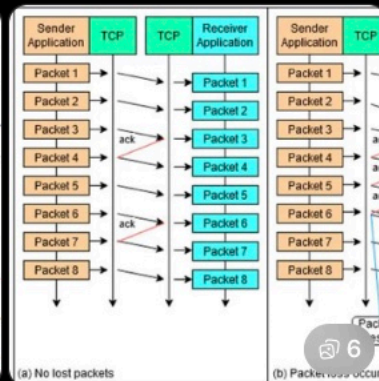
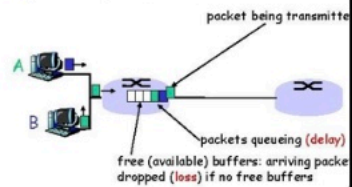
- Jitter buffer used to smooth variations

4. Packet Loss Rate

How do loss and delay occur

packets *queue* in router buffers

- packet arrival rate to link exceeds output capacity
- packets queue, wait for turn



Idea

- Percentage of packets **lost** during transmission

$$\text{Packet Loss Rate} = \frac{\text{Lost Packets}}{\text{Total Sent Packets}}$$

Causes

- Congestion (buffer overflow)
- Link errors
- Wireless interference

Example

- 100 packets sent, 5 lost → 5% loss

4. Packet Loss Rate



Idea

- Percentage of packets lost during transmission

$$\text{Packet Loss Rate} = \frac{\text{Lost Packets}}{\text{Total Sent Packets}}$$

Causes

- Congestion (buffer overflow)
- Link errors
- Wireless interference

Example

- 100 packets sent, 5 lost → 5% loss

⚡ Quick Comparison (Exam Ready)

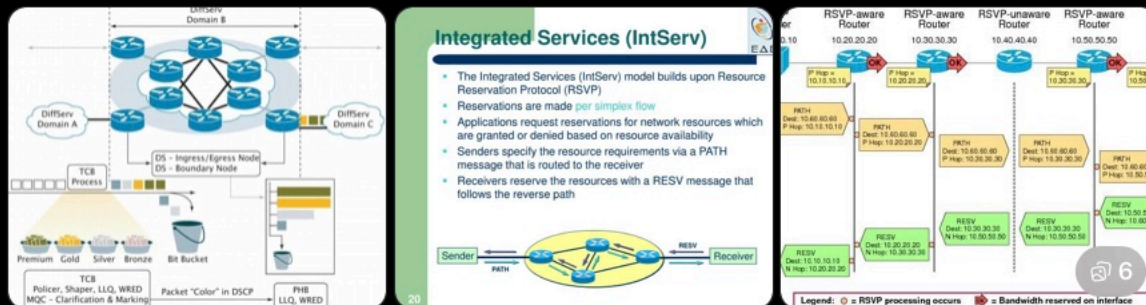
Metric	What it Measures	Ideal Value	Affects
Bandwidth	Data rate	High	Video quality
Delay	Time	Low	Gaming, calls
Jitter	Delay variation	Low	VoIP
Packet Loss	Reliability	Low	All apps

When we talk about **QoS in networks**, two classic architectures are used to guarantee or manage it:

👉 Integrated Services (IntServ) and Differentiated Services (DiffServ)

They solve the same problem (QoS), but in very different ways.

🧩 1. Integrated Services (IntServ)



◆ Core Idea

- Reserve resources for each flow
- Guarantees strict QoS (bandwidth, delay, etc.)

👉 Like booking a train seat in advance 🚂

◆ How it Works

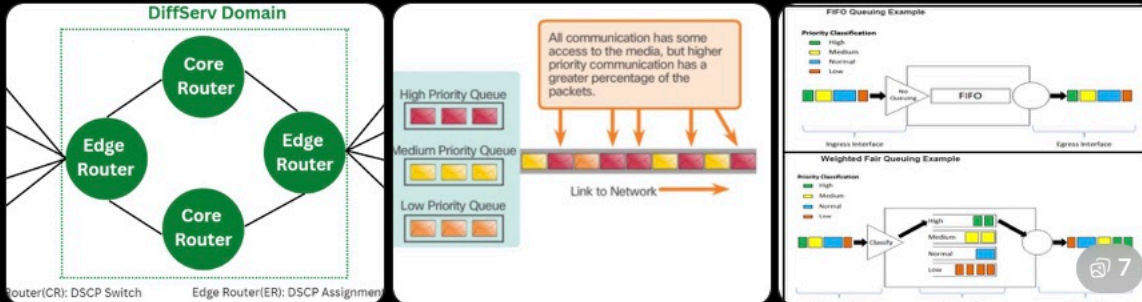
- Uses **RSVP**
- Steps:
 1. Sender sends request
 2. Network reserves bandwidth along path
 3. Routers maintain **per-flow state**

◆ Features

- ✓ Strong QoS guarantees
- ✓ Suitable for real-time apps (video conferencing)
- ✗ Not scalable (too many flows → too much state in routers)



2. Differentiated Services (DiffServ)



◆ Core Idea

- No reservation
- Packets are classified into priority classes

👉 Like VIP / normal / economy lanes on a highway 🚗

◆ How it Works

- Uses **DSCP (Differentiated Services Code Point)** in IP header
- Routers:
 - Don't store per-flow info
 - Just forward based on priority

◆ Features

- ✓ Scalable (works for Internet)
- ✓ Simple router design

✗ No strict guarantees (only relative priority)



IntServ vs DiffServ (GATE Gold Table)

Feature	IntServ	DiffServ
Approach	Per-flow	Per-class
Resource Reservation	Yes	No
Protocol	<u>RSVP</u>	DSCP marking
QoS Guarantee	Strict	Best effort with priority
Scalability	Poor	Excellent
Router State	High	Low

► 7.3 Integrated vs. Differentiated Services

Feature	IntServ (Integrated Services)	DiffServ (Differentiated Services)
Approach	Per-flow reservations	Aggregate class-based QoS
Protocol	RSVP (Resource reSerVation Protocol)	DSCP marking in IP header
Scalability	Poor (state per flow)	Good (state per class)
Guarantee	Hard QoS guarantees	Soft/relative guarantees
Use	ATM networks, small networks	Internet backbone, enterprise

MODULE 8: Wireless Networks & Mobility

► 8.1 Wireless Challenges

- **Hidden Terminal Problem** – A and C can't hear each other, both transmit to B → collision
- **Exposed Terminal Problem** – B hears A talking to C, unnecessarily avoids transmitting
- **Multipath Fading** – signals bounce off objects causing interference
- **Mobility** – IP address changes as device moves

🔑 **KEY:** RTS/CTS (Request to Send / Clear to Send) **solves** the Hidden Terminal Problem

► 8.2 MAC Protocols for Wireless

Protocol	Type	Key Feature
CSMA/CA	Contention-based	Avoid collisions with backoff + RTS/CTS
TDMA	Scheduled	Fixed time slots per device
OFDMA	Scheduled	Multiple subcarriers — used in LTE/ 5G
ALOHA	Random access	Simple, high collision probability
Slotted ALOHA	Random access	Better efficiency than pure ALOHA

► 8.3 Mobile IP

Mobile IP allows a device to keep the same IP address while moving between networks.

Home Agent (HA): Router at home network; maintains binding table

Foreign Agent (FA): Router at visited network; receives packets for mobile node

Care-of Address (CoA): Temporary IP at the visited network

KEY: **Triangular Routing Problem:** All packets go Source → HA → FA → Mobile (inefficient). Route Optimization solves this.

► 8.4 Ad-hoc Routing Protocols

◆ AODV – Ad hoc On-Demand Distance Vector

AODV is a reactive protocol — routes are only discovered when needed.

Step	Action	Message Used
1	Source broadcasts route request	RREQ (Route Request)
2	Intermediate nodes forward RREQ	RREQ
3	Destination or node with route replies	RREP (Route Reply)
4	Reverse path is established	RREP travels back
5	If link breaks , notify source	RERR (Route Error)
✓ Advantages	✗ Limitations	
On-demand — no route overhead when idle	Route discovery delay (latency)	
Supports mobile nodes well	Frequent RREQs in high-mobility scenarios	
Loop-free routes using sequence numbers	Not efficient for large-scale networks	

► 8.5 Group Communication & Multicast

Multicast delivers a single packet to a group of receivers simultaneously — more efficient than unicast to each.

Type	Description	Example
Unicast	1 sender → 1 receiver	HTTP file download
Broadcast	1 sender → all nodes	ARP request
Multicast	1 sender → group of receivers	IPTV, video conferencing
Anycast	1 sender → nearest group member	DNS root servers

IGMP: Internet Group Management Protocol — hosts join/leave multicast groups

PIM-SM: Protocol Independent Multicast — Sparse Mode; builds multicast trees

MODULE 9: Network Coding Techniques

► 9.1 What is Network Coding?

Traditional networks just forward packets. Network Coding allows intermediate nodes to combine (encode) packets before forwarding — increasing throughput!

KEY: Classic Example: Butterfly Network — XOR coding allows both A and B to receive each other's packets simultaneously through a shared bottleneck link.

Type	Description	Advantage
Linear Network Coding	Nodes create linear combinations of packets	Maximizes multicast throughput
Random Linear NC	Random coefficients — no coordination needed	Simpler to implement
XOR Coding	Simple bitwise XOR of packets	Low complexity, wireless networks
Regenerating Codes	For distributed storage — efficient repair	Reduces repair bandwidth
✓ Advantages	✗ Limitations	
Increases multicast throughput to max-flow bound	Increased computational complexity	
Reduces bandwidth in peer-to-peer networks	Decoding overhead at receiver	
Robust to link failures	Security risks (pollution attacks)	

MODULE 10: Overlay Networks

🌐 Network Coding (Simple + GATE Ready)

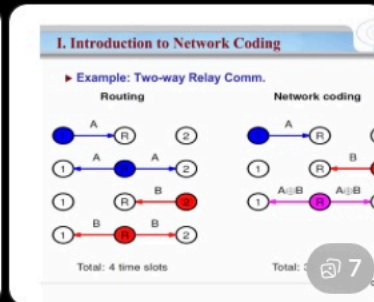
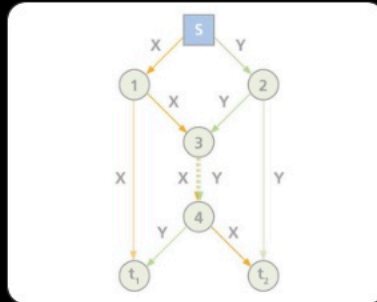
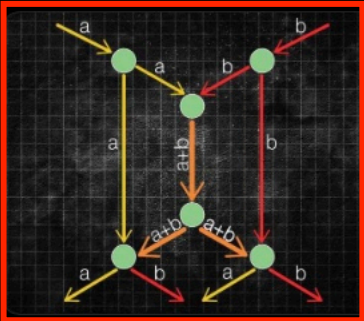
Network coding is a technique where **intermediate nodes don't just forward packets—they combine them** before forwarding.

- 👉 Traditional networking = **store & forward**
- 👉 Network coding = **mix (code) & forward**

💡 Core Idea (Intuition)

Instead of sending packets separately, nodes **encode multiple packets together** to improve efficiency.

📦 Classic Example (Butterfly Network)



💡 Without Network Coding

- Source sends A and B separately
- Bottleneck link must carry both → **more transmissions**

💡 With Network Coding

- Intermediate node sends:

$$A \oplus B$$

(XOR of A and B)

- Receivers decode:
 - If node has A → can get B
 - If node has B → can get A

👉 Same link, more information delivered!

💡 Key Operation

XOR-based coding (most common)

$$C = A \oplus B$$

- Receiver uses known packet to decode unknown one

◆ Types of Network Coding

1. Linear Network Coding

- Packets combined using linear algebra (XOR, addition, etc.)
- Most widely used

2. Random Linear Network Coding (RLNC)

- Packets combined with random coefficients
- Very powerful in dynamic networks

◆ Advantages

- ✓ Improves **throughput**
- ✓ Better **bandwidth utilization**
- ✓ Increases **robustness** (handles packet loss better)
- ✓ Efficient for **multicast**

◆ Disadvantages

- ✗ Extra computation (encoding/decoding)
- ✗ Complexity in implementation
- ✗ Delay due to coding/decoding

◆ Where It Is Used

- Peer-to-peer networks
- Wireless networks
- Satellite communication
- Content distribution systems

◆ Network Coding vs Traditional Routing

Feature	Traditional Routing	Network Coding
Operation	Forward packets	Combine packets
Efficiency	Lower	Higher
Complexity	Low	Higher
Throughput	Limited	Improved

Bottleneck Link (Computer) Free offer x

A **bottleneck link** is the **slowest link** in a **network path** that limits the overall data transfer rate.

👉 Think of it like:

A wide highway suddenly narrowing into a single lane — traffic slows down there.

◆ Core Idea

- Data travels through multiple links from source → destination
- Each link has a **capacity (bandwidth)**
- The link with the **minimum bandwidth** becomes the bottleneck

Visual Understanding



◆ Example

Suppose a path:

- Link 1 → 10 Mbps
- Link 2 → 2 Mbps ❌
- Link 3 → 8 Mbps

👉 End-to-end throughput = 2 Mbps

✓ Because the slowest link (Link 2) limits everything

◆ Important Formula

$$\text{Throughput} = \min(\text{All link bandwidths})$$

◆ Why It Matters

- Causes **congestion**
- Increases **delay**
- Reduces **overall performance**

◆ Real-Life Example

- Streaming video:
 - Your WiFi = 100 Mbps
 - ISP link = 5 Mbps ❌

👉 Video quality limited by 5 Mbps (bottleneck)

► 10.1 What is an Overlay Network?

An overlay network is built on top of the physical network using virtual links. Applications create their own 'virtual' network topology.

🔑 **KEY:** Examples: VPNs, P2P networks, CDNs, RON — all are overlay networks!

An overlay node is: A host (computer/server/VM) that runs a program which connects it to other nodes in a virtual network

► 10.2 RON – Resilient Overlay Network

RON routes around failures by probing multiple paths and choosing the best.

✓ Advantages	✗ Limitations
Detects and avoids failures in seconds (vs BGP minutes)	Scalability limited (N^2 probe cost)
Can optimize for latency or throughput	Requires overlay nodes at strategic points
Works above existing IP routing	Can't fix physical failures

► 10.3 P2P – Peer-to-Peer Networks

In P2P, every node is both client and server. No central authority needed.

P2P Type	How it Works	Example
Unstructured	Random connections, flooding for search	Gnutella, early Napster
Structured (DHT)	Distributed Hash Table for exact lookups	Chord, Kademlia, BitTorrent DHT
Hybrid	Central index + P2P data transfer	Old Napster, Spotify

DHT Key Idea: Each node is responsible for a portion of the key space. lookup(key) routes to the correct node in $O(\log N)$ hops.

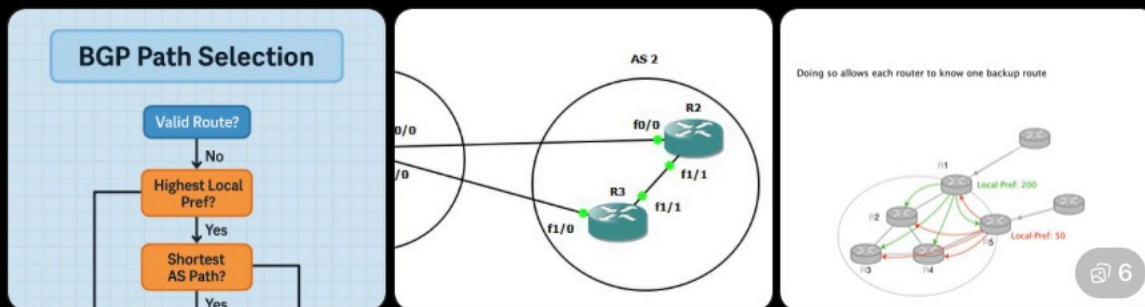
► 10.4 CDN – Content Delivery Network

CDNs replicate content across geographically distributed servers (edge nodes) to serve users from the nearest location.

Step	What Happens
1	User requests www.example.com
2	DNS resolves to nearest CDN edge server
3	If cached → serve immediately
4	If not cached → fetch from origin, cache, serve
5	User gets low-latency response
✓ Advantages	✗ Limitations

1. How BGP Fixes a Broken Path

(Used by ISPs across the Internet)



What happens when a link fails?

Example:

A → ISP1 → ISP2 → B ❌ (link breaks)



Step-by-step:

1. Router detects failure
2. Sends withdraw message (route no longer valid)
3. Updates propagate across Internet (AS to AS)
4. Each router recomputes best path using BGP rules:
 - Shortest AS path
 - Policies (business agreements)
5. New path is selected:

A → ISP3 → ISP4 → B ✅



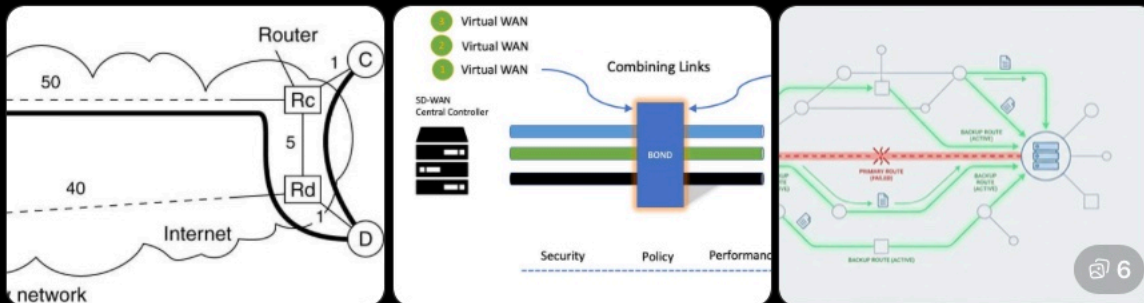
Problem with BGP

- Slow convergence:
 - Can take **seconds to minutes**
- Reasons:
 - Global coordination
 - Policy constraints (not just shortest path)
 - Loop prevention mechanisms

👉 BGP is **safe and scalable**, but **not fast**

⚡ 2. How RON Handles the Same Problem

(Overlay network at application layer)



Same failure:

A → ISP1 → ISP2 → B ❌



RON approach:

1. Overlay nodes (A, C, D...) **continuously monitor paths**
 - Delay
 - Loss
 - Throughput
2. A already knows:
 - A → C → B is working ✅
3. Immediately switches:

A → C → B ✅



👉 No waiting for Internet to “fix itself”

✂ Key Difference (THIS is the real answer)

Aspect	BGP	RON
Layer	Network layer	<u>Application layer</u>
Control	<u>ISPs</u>	End hosts
Knowledge	Limited (routing tables)	Measured performance
Goal	<u>Policy + reachability</u>	<u>Performance + reliability</u>
Reaction speed	Slow	Fast
Path choice	“Allowed” path	“Best observed” path

🌐 Peer-to-Peer (P2P) Networks — Easy + Technical

A **Peer-to-Peer (P2P)** network is a system where **every node (peer)** can act as both a **client** and a **server**.

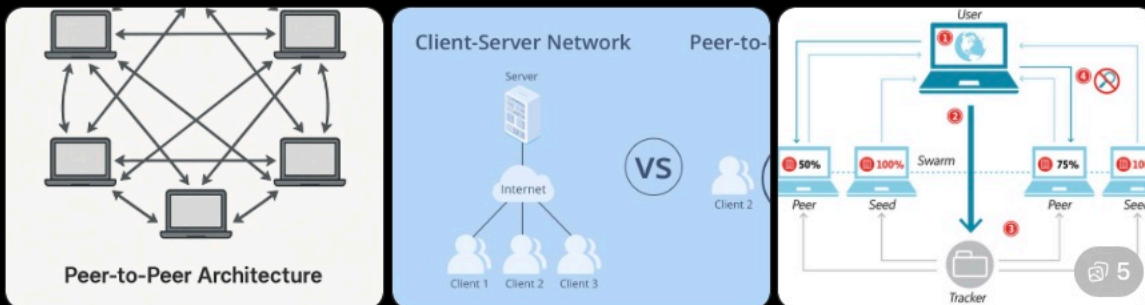
👉 No central server controlling everything.

💡 Core Idea

Each computer can **request data AND provide data**

- In traditional client-server:
 - Client → requests
 - Server → responds
- In P2P:
 - Every node = **client + server**

🖼️ Visual Understanding



💡 How It Works (Step-by-Step)

Let's take file sharing:

1. File is divided into **small pieces**
2. Different peers have different pieces
3. A new peer:
 - Downloads pieces from multiple peers
 - Uploads pieces to others

👉 This is called a **swarm**

Example system: [BitTorrent](#)

💡 Key Characteristics

- ✓ Decentralized
- ✓ Scalable
- ✓ Resource sharing (bandwidth, storage, CPU)

◆ Types of P2P Networks

🎁 Free offer ×

1. ◆ Unstructured P2P

- Random connections
- Example: early file sharing systems

✓ Flexible

✗ Inefficient search

2. ◆ Structured P2P

- Uses **Distributed Hash Table (DHT)**
- Organized lookup

✓ Fast search

✓ Efficient

3. ◆ Hybrid P2P

- Combines central server + P2P
- Example: Napster-like systems

◆ Advantages

✓ No single point of failure

✓ High scalability

✓ Efficient resource utilization

✓ Faster downloads (parallel downloads)

◆ Disadvantages

✗ Security issues

✗ Hard to manage/control

✗ Data reliability depends on peers

◆ P2P vs Client-Server

Feature	P2P	Client-Server
Control	Decentralized	Centralized
Scalability	High	Limited
Failure	No single point	Server failure critical
Cost	Low	High

◆ Real-World Applications

- File sharing → [BitTorrent](#)
- Blockchain networks
- VoIP systems (early Skype)



Step	What Happens
Reduced latency — serve from nearby node	Expensive to deploy globally
Reduces load on origin server	Cache invalidation complexity
High availability — if one node fails, redirect	Data consistency challenges

► 10.5 Web Caching

Web caches store copies of web content closer to users, reducing bandwidth and latency.

Cache Type	Location	Purpose
Browser Cache	Client browser	Store recently visited pages
Proxy Cache	ISP / Enterprise gateway	Shared cache for many users
Reverse Proxy	Near web server	Protect origin, serve popular content
CDN Edge	ISP's network edge	Global scale web caching

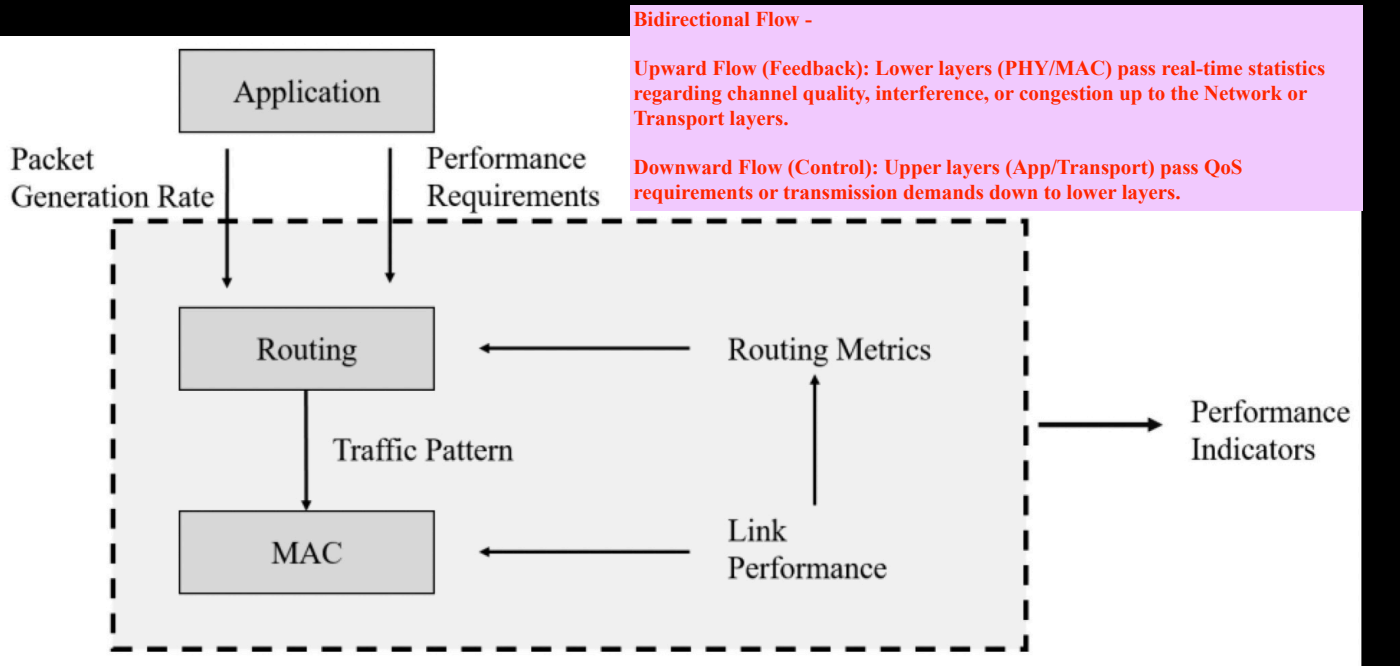
 **KEY:** HTTP cache headers: Cache-Control, Expires, ETag, If-Modified-Since control caching behavior.

► 10.6 Cross-Layer Optimization

Traditional layered design ignores interactions between layers. Cross-layer design allows layers to share information to optimize overall performance — especially in wireless networks.

Example: MAC layer tells TCP about wireless loss (so TCP doesn't think it's congestion). Result: better TCP performance on lossy wireless links.

✓ Advantages	✗ Limitations
Better performance by exploiting layer interactions	Breaks layered abstraction — harder to maintain
Critical for wireless where channel conditions change	Tight coupling makes systems fragile
Enables adaptive rate control	Non-standard: reduces interoperability



◆ What is "Cross-Layer Sensing"?

Using information from one layer to improve decisions at another layer

◆ Simple Example

Normal Approach (Layered ❌)

- Network layer chooses route
- MAC layer handles transmission
- Physical layer handles signal

👉 No sharing → inefficient

Cross-Layer Approach (✓)

Example:

- **Physical layer** detects:
 - Low signal strength ⚠️
- Shares with **Network layer**
- Network layer:
 - Chooses **different route**

👉 Result: better reliability + energy saving

Key Examples of Information

Flow: PHY to MAC/Network: Physical layer signals high congestion or noise, prompting the network layer to reroute traffic or the MAC layer to change transmission power.

MAC to Transport: Packet loss information helps the transport layer (e.g., TCP) understand it's a wireless link issue rather than congestion, preventing unnecessary, drastic rate reduction.

Application to PHY: High-definition video application notifies lower layers to prioritize bandwidth, prompting adaptive modulation

◆ Key Cross-Layer Interactions

1. Physical ↔ MAC

- Signal strength → affects retransmissions
- Channel condition → scheduling

2. MAC ↔ Network

- Congestion info → routing decisions
- Link quality → path selection

3. Application ↔ All Layers

- Data importance → priority handling
- Event detection → adaptive sampling

◆ Real Example (Energy Saving)

Sensor node behavior:

1. Battery low 🔋
2. Physical layer reports it
3. Network layer:
 - Avoids using this node in routing

👉 Extends network lifetime



MODULE 11: Emerging Network Types

► 11.1 Data Centre Networks

Data centres connect thousands of servers with ultra-low latency and high bandwidth.

Topology	Structure	Advantage
Fat Tree	Multi-level tree with redundant paths	Full bisection bandwidth
VL2	Valiant Load Balancing — random routing	Traffic agnostic, simple
BCube	Server-centric recursive structure	High fault tolerance
DCell	Recursive hypercube-like structure	Good scalability

 **KEY:** Data centre networks use Software Defined Networking (SDN) and ECMP (Equal-Cost Multi-Path) for load balancing.

► 11.2 DTN – Delay Tolerant Networks

DTNs work in environments where continuous end-to-end paths don't exist (space, remote areas, disaster zones).

Store-Carry-Forward: A node stores a message until it meets another node, carries it, then forwards. Like postal mail!

✓ Advantages	✗ Limitations
Works without continuous connectivity	Very high latency (hours to days)
Robust in extreme environments (space, military)	Requires large storage buffers
Messages eventually delivered	Complex routing (epidemic, spray-and-wait)

► 11.3 4G LTE & WiMAX

Feature	4G LTE	WiMAX (802.16)
Full Name	Long Term Evolution	Worldwide Interoperability for Microwave Access
Peak Speed	~100 Mbps (DL)	~70 Mbps (DL)
Access Tech	OFDMA + MIMO	OFDMA
Use Case	Mobile broadband	Fixed + mobile broadband
Architecture	EPC (Evolved Packet Core)	ASN Gateway + CSN
Handoff	Fast seamless handoff	Slower handoff

 **KEY:** LTE key components: eNodeB (base station), MME (mobility), SGW (data plane), PGW (internet gateway)

► 11.4 Online Social Networks (OSN)

OSNs (Facebook, Twitter, Instagram) create massive graph-structured networks connecting billions of users.

Property	Description	Example
Small World	Average path length is short (~6 hops)	6 degrees of separation
Power Law	Few nodes have many connections	Celebrity with millions of followers
Clustering	Friends of friends tend to be friends	Social communities
Community Structure	Naturally occurring groups	Interest-based groups

Challenge: Privacy, data sovereignty, fake news propagation, influence maximization — all are active OSN research areas.

► 11.5 WSN – Wireless Sensor Networks

WSNs consist of small, low-power sensors that collect environmental data (temperature, pressure, motion) and send it to a base station.

Component	Function
Sensor Node	Sense environment, process data, communicate
Gateway / Sink	Collect data from all sensors, connect to internet
Base Station	Process aggregated sensor data

Cross-Layer Sensing & Data Dissemination: WSNs use cross-layer protocols where physical layer info (signal strength, energy) affects MAC and routing decisions to maximize battery life.

✓ Advantages	✗ Limitations
No manual wiring needed	Severe energy constraints (battery-powered)
Self-organizing mesh topology	Limited processing and memory
Wide-area environmental monitoring	Unreliable wireless links

Dissemination - the action or fact of spreading something, especially information, widely.

MODULE 12: Emerging Applications

► 12.1 VoIP – Voice over IP

VoIP transmits voice calls over the internet instead of traditional phone networks. Examples: Skype, WhatsApp Calls, Zoom.

Delay (latency) is the total time it takes for data to travel from source to destination, while jitter is the variation or inconsistency in that delay over time.

Requirement	Acceptable Range	Why It Matters
Delay (One-Way)	< 150 ms	Conversations feel natural
Jitter	< 30 ms	Smooth audio playback
Packet Loss	< 1%	Intelligible voice quality
Bandwidth	8–64 kbps	Depends on codec (G.711, G.729)

Jitter Buffer: Playout buffer absorbs jitter — stores packets and plays them at steady rate. Too large = latency. Too small = underflow.

► 12.2 SIP – Session Initiation Protocol

SIP is the signaling protocol used to establish, modify, and terminate **multimedia sessions** (calls, video conferences).

SIP Message	Purpose
INVITE	Initiate a call session
200 OK	Call accepted
BYE	End the session
REGISTER	Register location with SIP server
ACK	Confirm final response

KEY: SIP **only does signaling**. Actual media (voice/video) is carried by **RTP (Real-time Transport Protocol)**.

► 12.3 Video over P2P

P2P video streaming distributes video delivery load across peers instead of a central server — like **BitTorrent** for live video.

System	Approach	Example
Tree-based	Peers form distribution tree	First-gen P2P TV
Mesh-based	Random peer connections, chunk exchange	PPLive, Joost
Hybrid CDN-P2P	CDN for popular content, P2P for rest	Akamai + P2P swarms
✓ Advantages	✗ Limitations	
Scales without expensive servers	Quality depends on peer availability	
Upload bandwidth of peers utilized	High churn = buffering and delays	
Cost-effective for provider	Legal and copyright issues	

Quick Revision — All Key Protocols at a Glance

Protocol	Layer	Full Name	Key Use
IP	Network	Internet Protocol	Packet routing
TCP	Transport	Transmission Control Protocol	Reliable data transfer
UDP	Transport	User Datagram Protocol	Fast, unreliable transfer
HTTP	Application	HyperText Transfer Protocol	Web browsing
DNS	Application	Domain Name System	Name → IP resolution
OSPF	Network	Open Shortest Path First	Link-state routing
BGP	Network	Border Gateway Protocol	Inter-AS routing
RSVP	Network	Resource Reservation Protocol	IntServ QoS
RTP	Application	Real-time Transport Protocol	Audio/Video streaming
SIP	Application	Session Initiation Protocol	VoIP call setup
AODV	Network	Ad hoc On-Demand Distance Vector	Wireless ad-hoc routing
IGMP	Network	Internet Group Mgmt Protocol	Multicast group management
SNMP	Application	Simple Network Mgmt Protocol	Network monitoring
ARP	Link/Network	Address Resolution Protocol	IP → MAC translation